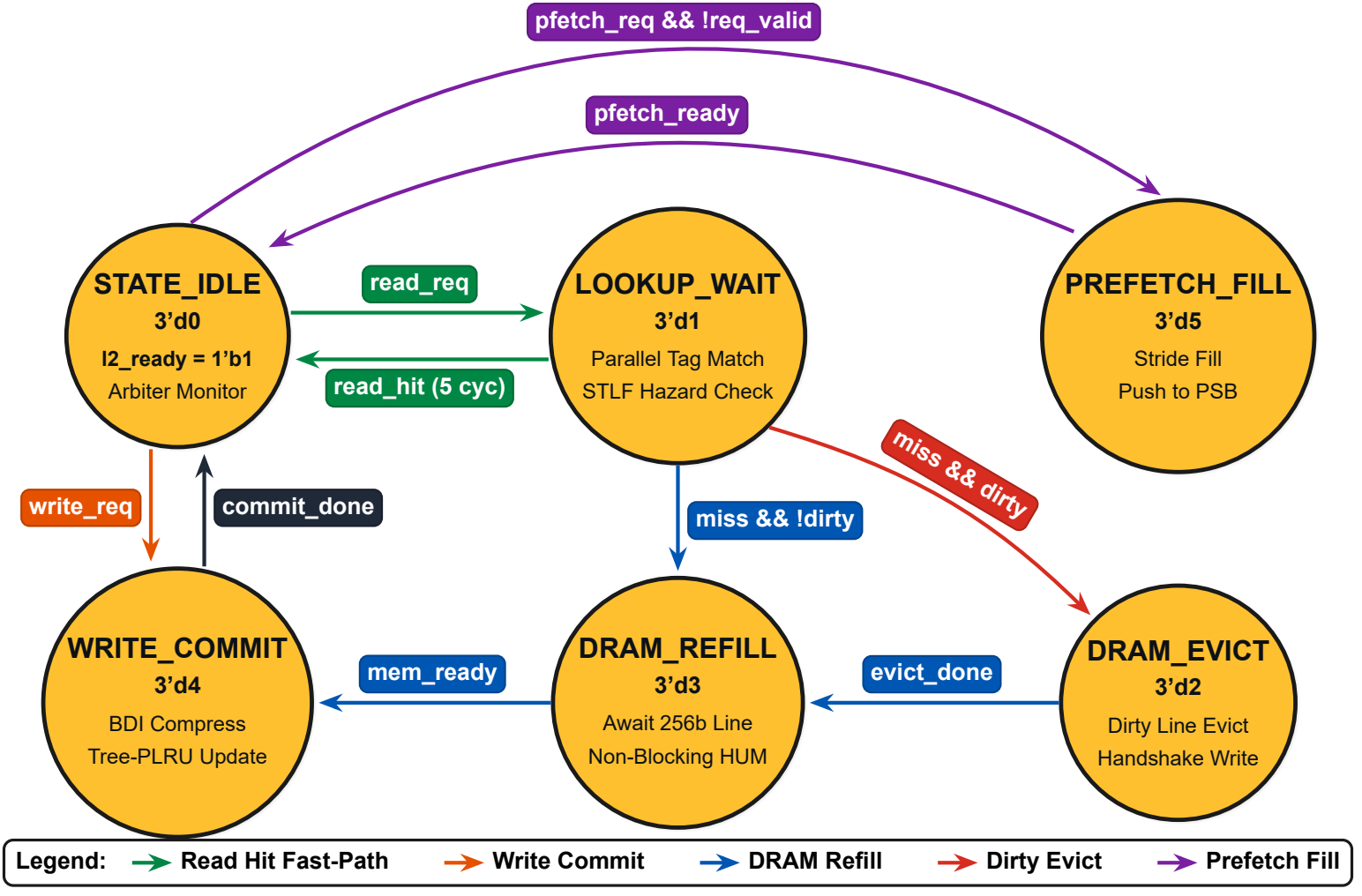

pfetch_req && !req_valid
pfetch_ready
STATE_IDLE
3'd0
l2_ready = 1'b1
Arbiter Monitor
read_req
read_hit (5 cyc)
LOOKUP_WAIT
3'd1
Parallel Tag Match
STLF Hazard Check
PREFETCH_FILL
3'd5
Stride Fill
Push to PSB
write_req
commit_done
miss && !dirty
miss && dirty
WRITE_COMMIT
3'd4
BDI Compress
Tree-PLRU Update
mem_ready
DRAM_REFILL
3'd3
Await 256b Line
Non-Blocking HUM
evict_done
DRAM_EVICT
3'd2
Dirty Line Evict
Handshake Write
Legend:
Read Hit Fast-Path
Write Commit
DRAM Refill
Dirty Evict
Prefetch Fill